

Assignment 3 Solution

Q.1

$\vec{x}^{(1)} = [1 \ -1 \ 1]^T, \ y^{(1)} = ? \quad k = 3$

i	d_i (Euclidean)	d_i (Manhattan)	Rank (Euclidean)	Rank (Manhattan)
1	2.0	2.0	1	1
2	2.828	4.0	4	3
3	4.242	6.0	7	4
4	2.443	4.0	3	3
5	3.741	6.0	6	4
6	3.162	4.0	5	3
7	3.162	4.0	5	3
8	3.741	6.0	6	4
9	2.236	3.0	2	2

Nearest three points. Labels

Euclidean	Manhattan
1, 1, 1	1, 1, -1 or +1
majority voting	majority voting
Euclidean	Manhattan
1	1
$\therefore$ label $y^{(1)} = 1$	$y^{(1)} = 1$

$$\bar{x}^{(2)} = [-2 \ 3 \ -1]^T, \quad y^{(2)} = ? \quad K=3$$

i	d_i (Euclidean)	d_i (Manhattan)	Rank (Euclidean)	Rank (Manhattan)
1	3.605	5.0	4	2
2	2.236	3.0	1	1
3	2.236	3.0	1	1
4	5.0	7.0	8	4
5	3.0	3.0	2	1
6	4.582	7.0	7	4
7	4.358	7.0	5	4
8	3.316	5.0	3	2
9	4.472	6.0	6	3

Nearest three points labels

Euclidean

-1, -1, -1

majority voting

Euclidean

-1

$\therefore$ Label $y^{(2)} = -1$

Manhattan

-1, -1, -1

majority voting

Manhattan

-1

Label $y^{(2)} = -1$

Q.2

$$\bar{x}^{(1)} = [1 \ -1 \ 1]^T$$

$$y^{(1)} = ?$$

Nearest three points

Labels

Euclidean

Manhattan

2.9, 6.1, 8.9

2.9, 6.1, 3.1/8.5/9.0/8.7

$\therefore$ Avg. label value

$\therefore$ Avg. label value

Euclidean 5.966

4.033/5.966/6.0/5.899

$$\therefore y^{(1)} = 5.966$$

$$\therefore y^{(1)} = 4.033 \text{ or } 5.966 \text{ or } 6.0 \text{ or } 5.899$$

$$\bar{x}^{(2)} = [-2 \ 1 \ -1]^T, \quad y^{(2)} = ?$$

Nearest three points

Labels

Euclidean

Manhattan

3.1, 8.8, 8.6

3.1, 8.8, 8.6

$\therefore$ Avg. label value

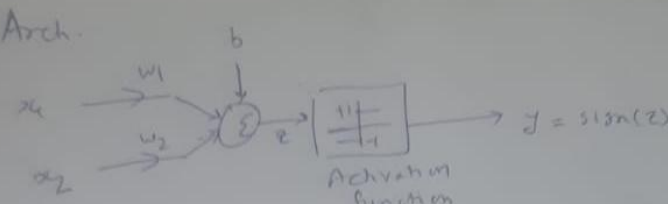
$\therefore$ Avg. label value

Euclidean 6.833

Manhattan: 6.833

Q.3

Arch.



$$z = \bar{w}^T \bar{x} + b$$

$$\bar{w} = [w_1, w_2]^T, \quad \bar{x} = [x_1, x_2]^T$$

model eqⁿ: $y = \text{sign}(\bar{w}^T \bar{x} + b)$

$$= \begin{cases} +1 & ; \bar{w}^T \bar{x} + b > 0 \\ -1 & ; \bar{w}^T \bar{x} + b \leq 0 \end{cases}$$

Given $\bar{w}^* = [-0.10, 0.15]$, $b^* = 0.27$

$$\therefore z = -0.10x_1 + 0.15x_2 + 0.27$$

- $\bar{x}^{(1)} = [-0.17, 0.51]$, $y^{(1)} = -1$

$$\therefore z^{(1)} = 0.3839$$

$$\therefore y^{(1)} (\text{Predicted}) = \text{sign}(z^{(1)}) = \text{sign}(0.3839) = 1 \neq y^{(1)}$$

$\therefore$ misclassified.

- $\bar{x}^{(2)} = [0.85, 0.51]$, $y^{(2)} = +1$

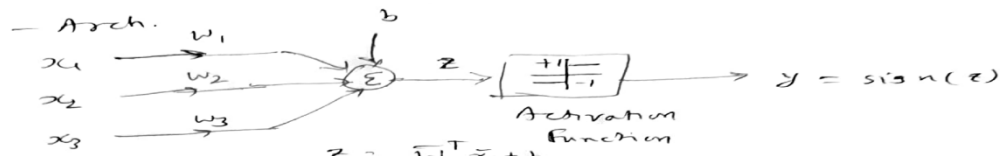
$$\therefore z^{(2)} = 0.2839$$

$$\therefore y^{(2)} (\text{Predicted}) = \text{sign}(z^{(2)}) = \text{sign}(0.2839) = 1 = y^{(2)}$$

$\therefore$ correctly classified

$\therefore$ no. of misclassifications = 1

Q.4



$$\bar{w} = [w_1 \ w_2 \ w_3]^T, \quad \bar{x} = [x_1 \ x_2 \ x_3]^T \quad (2)$$

- Model Eqn: $y = \text{sign}(\bar{w}^T \bar{x} + b)$

$$= \begin{cases} 1; & \bar{w}^T \bar{x} + b > 0 \\ -1; & \bar{w}^T \bar{x} + b \leq 0 \end{cases} \quad (2)$$

- $\bar{w}^* = [-0.1 \ 0.15 \ 0.27]^T$; $b^* = 0.27$

$$z = -0.1x_1 + 0.15x_2 + 0.27x_3 + 0.27$$

- $\bar{x}^{(1)} = [-0.17 \ 0.51 \ 0.85]^T$ $y^{(1)} = -1$

$$\therefore z^{(1)} = 0.6134 \quad \therefore y^{(1)}(\text{predicted}) = \text{sign}(0.6134) = +1 \neq y^{(1)}$$

- $\bar{x}^{(2)} = [0.85 \ 0.17 \ 0.47]^T$, $y^{(2)} = +1$

$$\therefore z^{(2)} = 0.226 \quad \therefore y^{(2)}(\text{predicted}) = \text{sign}(0.226) = +1 = y^{(2)}$$

$\therefore$ No. of mis-classⁿ errors = 1 (out of 2) (2)

Assignment 4 Solution

Q 1

For Logistic Regression binary classification method write the objective function. Find it's gradient. Write weight update rule using GD algorithm. Assume $\eta=1$. Obtain weight update for following dataset if all weights are initialized to 0. Labeled data set: $\{([-0.2, 0.5]^T, -1); ([0.2, -0.5]^T, +1)\}$

Logistic Regression objective function: (maximize log-likelihood)

$$\text{minimize } E(\bar{w}) = -\ln(L(\bar{w})) ; \ln(L(\bar{w})) = \sum_{i=1}^N \ln(L(y_i \bar{w}^T \bar{x}_i))$$

$$\text{Gradient } \frac{\partial E(\bar{w})}{\partial \bar{w}} = - \frac{\partial}{\partial \bar{w}} (\ln(L(\bar{w})))$$

$$= - \sum_{i=1}^N y_i (1 - L(y_i \bar{w}^T \bar{x}_i)) \bar{x}_i$$

Given $N=2$, $\bar{x}_1 = [-0.2, 0.5]^T$, $y_1 = -1$
 $\bar{x}_2 = [0.2, -0.5]^T$, $y_2 = 1$, $\bar{w}^{(0)} = [0, 0]^T$

$$\bar{w}^{(n+1)} = \bar{w}^{(n)} + \eta \frac{\partial E(\bar{w})}{\partial \bar{w}}^{(n)} \rightarrow \text{weight update rule } (\eta=1)$$

$$\bar{w}^{(1)} = \bar{w}^{(0)} - \left[\frac{\partial E(\bar{w})}{\partial \bar{w}} \right]^{(0)}$$

$$= [0, 0]^T + \sum_{i=1}^2 y_i (1 - L(0)) \bar{x}_i$$

$$= \cancel{[0, 0]^T} + y_1 \left(\frac{1}{2}\right) \bar{x}_1 + y_2 \left(\frac{1}{2}\right) \bar{x}_2 = \frac{1}{2} [-0.2, -0.5]^T + \frac{1}{2} [0.2, -0.5]^T$$

$$= [0.2, -0.5]^T$$

$L(x) = \frac{1}{1+e^{-x}}$
 $L(0) = \frac{1}{2}$

Q 2

For MCE binary classification method write the objective function. Find it's gradient. Write weight update rule using GD algorithm. Assume $\eta=1$. Obtain weight update for following dataset if all weights are initialized to 0. Labeled data set: $\{([-0.2, 0.5]^T, -1); ([0.2, -0.5]^T, +1)\}$

MSE objective function: (Reduce sum of classification error)

$$E(\bar{w}) = \sum_{i=1}^N \ell(-y_i \bar{w}^T \bar{x}_i) \quad (1)$$

Gradient $\nabla E(\bar{w}) = \frac{\partial E(\bar{w})}{\partial \bar{w}} = \sum_{i=1}^N -y_i \ell(y_i \bar{w}^T \bar{x}_i) (1 - \ell(y_i \bar{w}^T \bar{x}_i)) \bar{x}_i \quad (2)$

Weight update rule: (GD with $\eta = 1$)

$$\bar{w}^{(n+1)} = \bar{w}^{(n)} - \frac{\partial E(\bar{w})}{\partial \bar{w}} = \bar{w}^{(n)} + \sum_{i=1}^N y_i \ell(y_i \bar{w}^{(n)T} \bar{x}_i) (1 - \ell(y_i \bar{w}^{(n)T} \bar{x}_i)) \bar{x}_i \quad (3)$$

$\bar{w}^{(0)} = [0, 0]^T$; $N = 2$; $\bar{x}_1 = \begin{bmatrix} -0.2 \\ 0.5 \end{bmatrix}$, $y_1 = -1$
 $\bar{w}^{(1)} = \sum_{i=1}^2 y_i \ell(y_i \bar{w}^{(0)T} \bar{x}_i) (1 - \ell(y_i \bar{w}^{(0)T} \bar{x}_i)) \bar{x}_i$
 $\bar{x}_2 = \begin{bmatrix} -0.2 \\ -0.5 \end{bmatrix}$, $y_2 = +1$
 $\ell(x) = \frac{1}{1 + e^x}$
 $\ell(0) = \frac{1}{2}$
 $= -1 \left(\frac{1}{2}\right) \left(1 - \frac{1}{2}\right) \begin{bmatrix} -0.2 \\ 0.5 \end{bmatrix} + 1 \left(\frac{1}{2}\right) \left(1 - \frac{1}{2}\right) \begin{bmatrix} -0.2 \\ -0.5 \end{bmatrix}$
 $= \begin{bmatrix} 0.05 \\ -0.125 \end{bmatrix} + \begin{bmatrix} -0.05 \\ -0.125 \end{bmatrix} = \begin{bmatrix} 0.05 \\ -0.25 \end{bmatrix} \quad (3)$

Q 3 . Consider following labelled dataset for linear regression problem.

- $\mathbf{X} = \begin{bmatrix} -0.2 & 0.5 \\ 0.8 & 0.6 \\ 0.1 & -0.1 \end{bmatrix}$ and $\mathbf{y} = \begin{bmatrix} -0.1 \\ 0.4 \\ 0.2 \end{bmatrix}$
- It is to be fitted on linear model $y = \mathbf{w}^T \mathbf{x}$. Identify number of samples and number of features. Find optimum weight vector if we use Linear Regression.

Q. 2

$$\bar{\mathbf{X}} = \begin{bmatrix} -0.2 & 0.5 \\ 0.8 & 0.6 \\ 0.1 & -0.1 \end{bmatrix} \quad \bar{\mathbf{y}} = \begin{bmatrix} -0.1 \\ 0.4 \\ 0.2 \end{bmatrix} \quad (1)$$

$\therefore N = 3, d = 2$

Linear Regression:

$$\bar{\mathbf{w}}^* = (\bar{\mathbf{X}}^T \bar{\mathbf{X}})^{-1} (\bar{\mathbf{X}}^T \bar{\mathbf{y}}) \quad (2)$$

$$\bar{\mathbf{X}}^T \bar{\mathbf{X}} = \begin{bmatrix} -0.2 & 0.5 \\ 0.8 & 0.6 \\ 0.1 & -0.1 \end{bmatrix}^T \begin{bmatrix} -0.2 & 0.5 \\ 0.8 & 0.6 \\ 0.1 & -0.1 \end{bmatrix} = \begin{bmatrix} 0.69 & 0.37 \\ 0.37 & 0.62 \end{bmatrix}$$

$$\bar{\mathbf{w}}^* = \begin{bmatrix} 2.13 & -1.27 \\ -1.27 & 2.77 \end{bmatrix}^{-1} \begin{bmatrix} 0.36 \\ 0.77 \end{bmatrix} = \begin{bmatrix} 0.55 \\ -0.05 \end{bmatrix} \quad (3)$$

Q 4 Consider following labelled dataset for regression problem.

- $\mathbf{X} = \begin{bmatrix} -0.2 & 0.5 \\ 0.8 & 0.6 \\ 0.1 & -0.1 \end{bmatrix}$ and $\mathbf{y} = \begin{bmatrix} -0.1 \\ 0.4 \\ 0.2 \end{bmatrix}$
- Identify number of samples and number of features. It is to be fitted on linear model $y = \mathbf{w}^T \mathbf{x}$. Find optimum weight vector if we use **Ridge regression** with hyper-parameter value of 0.01.

$$\text{No. of samples} = 03 = N \quad - (1)$$

$$\text{No. of features} = 02 = d \quad \therefore \bar{w} \in \mathbb{R}^2 \quad - (1)$$

$$\lambda = 0.01 \quad - (1)$$

$$\bar{w}^* = (\bar{X}^T \bar{X} + \lambda \bar{I})^{-1} (\bar{X}^T \bar{y})$$

$$= \left[\begin{bmatrix} -0.2 & 0.8 & 0.1 \\ 0.5 & 0.6 & -0.1 \end{bmatrix} \begin{bmatrix} -0.2 & 0.5 \\ 0.8 & 0.6 \\ 0.1 & -0.1 \end{bmatrix} + \begin{bmatrix} 0.01 & 0 \\ 0 & 0.01 \end{bmatrix} \right]^{-1} \left[\begin{bmatrix} -0.2 & 0.8 & 0.1 \\ 0.5 & 0.6 & 0.1 \end{bmatrix} \begin{bmatrix} -0.1 \\ 0.4 \\ 0.2 \end{bmatrix} \right]$$

$$= \begin{bmatrix} 0.539 \\ -0.047 \end{bmatrix}$$

- (3)